

Agile + DevOps + Cloud Implementation Roadmap for University MIS

1. Project Vision

Deliver a **cloud-based, modular, and scalable University MIS** that streamlines all academic and administrative processes, supports real-time data access, integrates with third-party systems, and is continuously improved through Agile development and DevOps automation.

2. Agile Approach

2.1 Product Backlog (High-Level Features)

Epic 1 – Student Lifecycle Management

- Admissions & Enrolment Module
- Student Profile & Academic Records
- Course Registration & Timetabling
- Attendance Tracking

Epic 2 – Academic Administration

- Faculty Management
- Curriculum & Syllabus Management
- Examination & Grading System
- Digital Transcripts & Certificates

Epic 3 – Financial & Resource Management

- Fee Collection & Online Payments
- Hostel & Accommodation Management
- Library Management System
- Procurement & Inventory Management

Epic 4 – Communication & Collaboration

- Notifications (Email/SMS/App Push)
- Student-Faculty Portal & Chat

- Event Management & Calendar Integration

Epic 5 – Analytics & Compliance

- Academic Performance Dashboards
 - Financial Reports
 - Accreditation & Government Reporting
 - AI-based Early Warning for At-Risk Students
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2.2 Agile Execution Plan (Scrum Framework)

Sprint Length: 2 weeks

Roles:

- Product Owner – University IT Director
- Scrum Master – Project Manager
- Dev Team – Developers, QA Engineers, UX Designers, DevOps Engineers
- Stakeholders – Faculty, Admin Staff, Students

Sample Sprint Plan:

- **Sprint 1:** Setup cloud infrastructure, CI/CD pipeline, user authentication, role-based access.
 - **Sprint 2:** Student profile management + admissions module (basic version).
 - **Sprint 3:** Course registration, timetable management.
 - **Sprint 4:** Fee collection, payment gateway integration.
 - **Sprint 5:** Examination system, grading, results publishing.
 - **Sprint 6:** Library & hostel management integration.
 - **Sprint 7:** Analytics dashboard, reporting, and AI alerts.
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3. DevOps Implementation

3.1 CI/CD Pipeline

- **Source Code Management:** GitHub / GitLab
- **CI Tool:** Jenkins / GitHub Actions

- **Build & Test:** Automated unit, integration, and UI tests using PyTest, Selenium, and API testing tools
 - **Artifact Repository:** Nexus / AWS CodeArtifact
 - **Deployment Automation:** Ansible / Terraform
 - **Containerization:** Docker for packaging services
 - **Orchestration:** Kubernetes (EKS / AKS / GKE) for scaling and self-healing
 - **Monitoring & Logging:** Prometheus + Grafana + ELK Stack
 - **Security:** SAST/DAST (SonarQube, OWASP ZAP), secrets management with Vault
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3.2 DevOps Workflow

1. **Code Commit:** Developer pushes code to repository.
 2. **Build Trigger:** CI pipeline runs unit tests, builds Docker images.
 3. **Security Scan:** Static & dynamic security tests run automatically.
 4. **Deploy to Staging:** Test environment updated for QA review.
 5. **Automated Tests:** Integration, regression, and UI tests run.
 6. **Manual Approval:** Product Owner validates release readiness.
 7. **Production Deployment:** CD pipeline deploys to live cloud environment.
 8. **Monitoring & Feedback:** Application performance and errors tracked in real-time.
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4. Cloud Architecture

4.1 Proposed Stack

- **Cloud Provider:** AWS / Azure / GCP
- **Frontend:** React.js / Angular (served via CloudFront / Azure CDN)
- **Backend:** Node.js / Django / Spring Boot (containerized)
- **Database:** PostgreSQL / MySQL (RDS / Cloud SQL) + Redis Cache
- **Storage:** Amazon S3 / Azure Blob for documents & media
- **Authentication:** OAuth 2.0 / SAML, integrated with University SSO
- **APIs:** REST + GraphQL for integration with LMS, ERP, and government portals

4.2 Cloud Deployment Diagram

User → CDN → Load Balancer → App Services (K8s) → Database (RDS)

 ↳ Storage (S3)

 ↳ API Gateway → External Services (LMS, Payments)

 ↳ Analytics Service → BI Dashboard

5. Key Deliverables

- Fully functional **modular MIS** accessible via web and mobile.
 - Automated CI/CD pipeline ensuring **weekly releases**.
 - **Cloud-native deployment** with 99.9% uptime SLA.
 - **Role-based dashboards** for students, faculty, and administrators.
 - Real-time **data analytics & reports** for decision-making.
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6. Expected Benefits

- **Faster Feature Delivery:** From months to weeks.
- **Reduced Downtime:** Zero-downtime deployments via rolling updates.
- **Improved Accessibility:** Secure access from anywhere, any device.
- **Scalability:** Easily handle growing student and course volumes.
- **Cost Optimization:** Pay-as-you-go cloud model.